

Introduction to QGIS



Catalina Moreno
Jessica Benner

[Link](#)



GIS is a unique problem-solving technique

That use **Maps** to visualize and analyze
data

West Africa Region



Nigeria

Africa -World Map



Atlantic Ocean

This workshop
aims you can
recreate this
map!

And start thinking
in geospatial
relationships...

0

What do you need? **A starting point**



1

Problem



2

Tool



3

Data



4

Analyze
and



5

Story+
Print

1. A problem: A dam overflow in Nigeria the last week

Dam overflow sparks new crisis in insurgency-hit Nigerian city



Houses are partially submerged following a dam collapse in Maiduguri, Nigeria, Sept. 10, 2024.

<https://www.voanews.com/a/dam-overflow-sparks-new-crisis-in-insurgency-hit-nigerian-city/7778990.html>

Worst in 30 years

The United Nations refugee agency in Nigeria (UNHCR) on its X account said it was the city's worst flooding in 30 years.

"It is an unprecedented incident," NEMA spokesman Ezekiel Manzo told *AFP* on Tuesday. "Some of the central parts of the city that have not witnessed flood in so many years are witnessing it today."

READ MORE [Sudan floods: At least 132 killed this year, health ministry says](#)

Thousands of homes have been submerged by the **rapid rise of waters** after the rupture of the Alau dam on the Ngadda River, 20km (12 miles) south of Maiduguri.

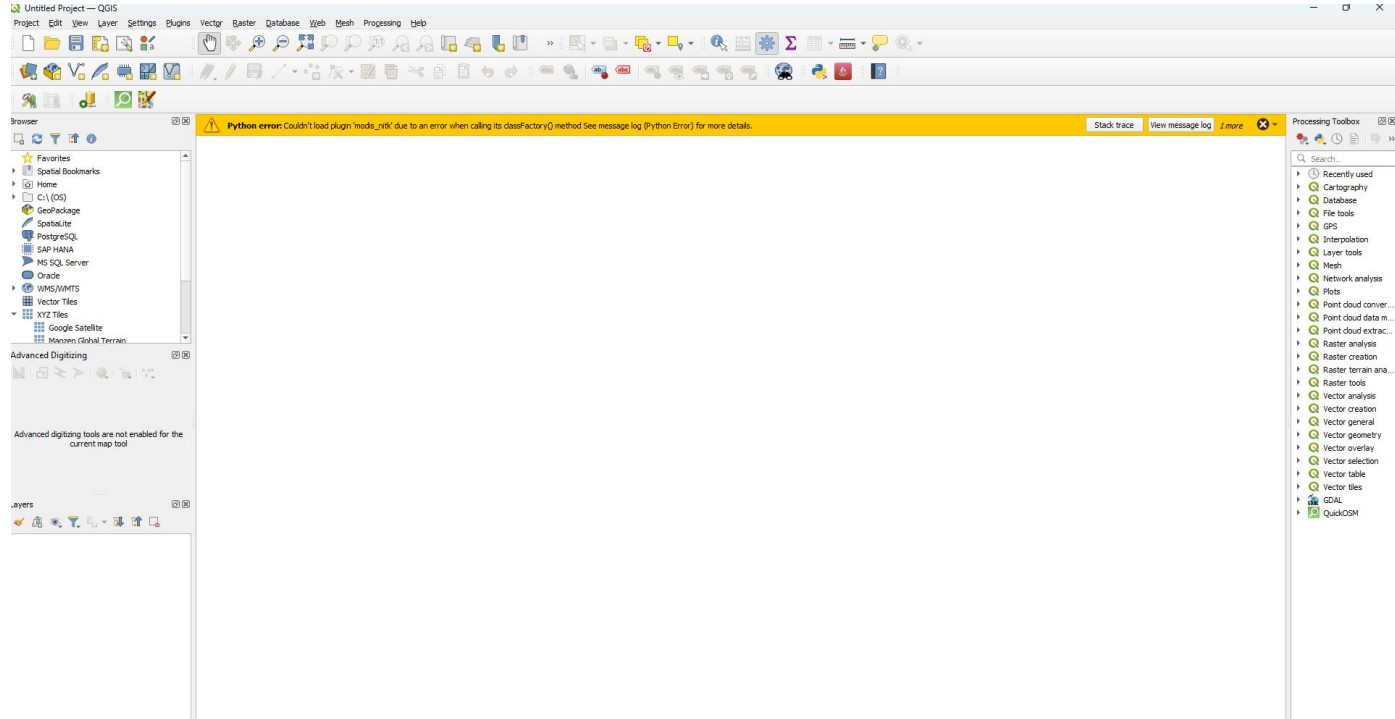
"The last three days have over 150,000 individuals with over 23,000 households affected," said NEMA zonal coordinator Surajo Garba.

But with more locations being hit, "we are sure the figure will be much over 200,000 individuals," Garba forecast.

Analyze How big is this crisis..

2. Tool: Download and set- up the QGIS

<https://qgis.org/>

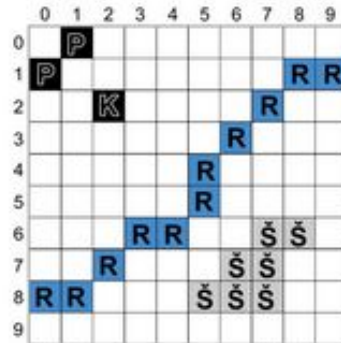


3. Data: Find your data

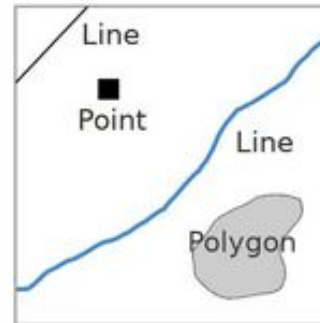


Real world

Raster view

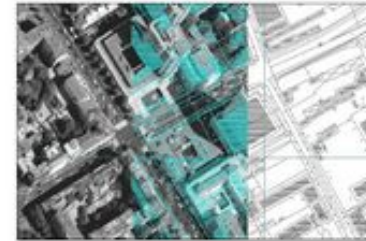


Vector view



Vektor format

Raster format



Tabular data

Address	Built	Square footage	Number of bedrooms
816 High Street	1888	2200	4 bedrooms

3. Data set & guide

Your QR Code



SCAN ME



GUIDE TO NIGERIA DAM OVERFLOW EXAMPLE: Q GIS WORKSHOP

 This step-by-step guide for the hands-on project in Nigeria, part of the "Introduction to QGIS" workshop, offers a practical introduction to using QGIS for mapping and basic spatial analysis.

 [Full Abstract](#)

 [Workshop_Slides.pdf](#) 1217.2KB

 [Workshop DATA](#)

3. Data: Find your data

Data

Type

Source

Political Boundaries

Shapefile (SHP)

www.gadm.org

Nigeria Population

Shapefile (SHP)

GHSL - Global Human Settlement Layer

Google Maps
/Open street maps

Server Connection

Server connection

Dam list

Table (CSV)

[FAO](#)

Rivers

Shapefile (SHP)

<https://data.humdata.org/>

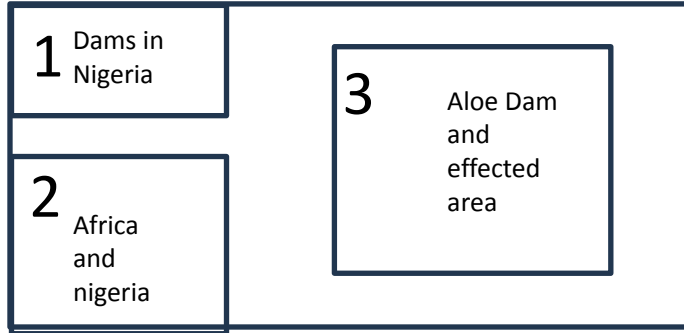
3. Analyze: but first Set Up a CRS



Taken from: opennews.org

4. MAPPING: Tell a story and Do some art!

Sketch it



MAP

Find a size

Zoom

Embedded Maps

Details

Scale
North

Title
Legent

Print it

SVG

PDF

So, lets start in our hands-up project!!



Thank you!

The library has **GIS SERVICES** to help you with
your projects

Survey

XYZ tiles servers

Google Maps

<https://mt1.google.com/vt/lyrs=r&x={x}&y={y}&z={z}>

Google Satellite

<https://www.google.cn/maps/vt?lyrs=s@189&gl=cn&x={x}&y={y}&z={z}>

Google Satellite Hybrid

<https://mt1.google.com/vt/lyrs=y&x={x}&y={y}&z={z}>

Google Terrain

<https://mt1.google.com/vt/lyrs=t&x={x}&y={y}&z={z}>

Google Roads

<https://mt1.google.com/vt/lyrs=h&x={x}&y={y}&z={z}>

Open street Maps

<https://mt1.google.com/vt/lyrs=h&x={x}&y={y}&z={z}>

3. **Analyze:** Work with your data

With the **shapefile**

- Filter (SQL)
- Properties

With the **csv**

- Upload data
- Properties

With **Base Map**

- Connect to server
- Search With Plugin

Carnegie
Mellon
University